

2nd Semester MCA Semester End Examination March 2026

Course Title: Computer Networks

Course Code: M21DE0203 - 86823

Duration: 3 Hours

Max. Marks: 100

Note:

1. Answer the required number of questions from **Section A to C**.
2. Verify and ensure that question paper is completely printed before answering the question paper.
3. Any queries/discrepancies regarding the question paper, must be brought to the notice of the invigilator.
4. **Students must check the course title and course code before answering the question paper.**

Section – A		Marks	CO	PO	BL
Answer any FIVE of the following questions carrying 2 marks each		5x2=10			
1.	Define Data Communication.		1	1	1
2.	Mention any two factors that affect network throughput.		2	2	1
3.	What is a protocol? Give any two examples of Application layer protocols.		1	2	2
4.	Mention the types of multiple access protocols.		3	2	1
5.	Mention the difference between TCP and UDP		4	2	2
6.	What is meant by Random access.		3	3	2
Section – B					
Answer any EIGHT of the following questions carrying 5 marks each		8x5=40			
7.	Differentiate between broadcast link and point-to-point link transmission technologies.		1	2	4
8.	Explain LAN, MAN, and WAN with suitable examples.		1	3	2
9.	Explain different types of transmission impairment.		2	3	2
10.	Differentiate between serial transmission and parallel transmission.		2	4	4
11.	Describe any two types of switching techniques.		2	4	2
12.	Differentiate between flow control and error control.		3	3	4
13.	Illustrate the Stop-and-Wait protocol using a diagram.		3	4	2
14.	Explain process-to-process delivery in the Transport Layer.		4	4	2
15.	Describe the working of ARP with a neat diagram.		4	3	2
16.	Explain the working of FDMA with a neat diagram.		3	2	2
Section – C					
Answer FIVE of the following questions carrying 10 marks each		5x10=50			
17.	Describe the OSI Reference Model with neat diagram and explain the functions of each layer		1	2	2
18.	Explain the different types of guided transmission media		2	2	2
19.	Compare different topologies with advantages and disadvantages.		1	3	4
20.	A data word 1101011011 is transmitted using CRC with generator 1001, Find the CRC.		3	4	4
21.	Explain Link State Routing and Distance Vector Routing algorithms in detail.		4	3	2
22.	What is congestion? Explain Leaky Bucket and Token Bucket algorithms.		4	4	2

2nd Semester MCA Semester End Examination March 2026

Course Title: Machine Learning

Course Code: M23DE0201 - 48623201

Duration: 3 Hours

Max. Marks: 100

Note:

1. Answer the required number of questions from **Section A to C**.
2. Verify and ensure that question paper is completely printed before answering the question paper.
3. Any queries/discrepancies regarding the question paper, must be brought to the notice of the invigilator.
4. **Students must check the course title and course code before answering the question paper.**

Section - A		Marks	CO	PO	BL																																												
Answer any FIVE of the following questions carrying 2 marks each		5x2=10																																															
1.	Justify the need of Semi supervised learning in Machine Learning.		1	1,3	2																																												
2.	What is overfitting ? How can it be prevented?		2	2,4	2																																												
3.	What is the purpose of activation functions in neural networks?		2	2,4	1																																												
4.	Discuss the importance of data preprocessing before model building.		3	1,4	1																																												
5.	Write about policy search with respect to reinforcement learning.		3	1,4	1																																												
6.	What is the significance of scaling data in machine learning models?		4	2,4	1																																												
Section - B																																																	
Answer any EIGHT of the following questions carrying 5 marks each		8x5=40																																															
7.	Explain Support Vector Machine (SVM) with soft margin and non-linear classification. How do kernel functions help in handling non-linearly separable data?		1	1,3	2																																												
8.	Explain Min- Max normalization and Z - score normalization with examples.		1	1,3	2																																												
9.	What is Hyperparameter ? Discuss in detail the fine tuning of hyperparameters in neural networks .		2	2,4	2																																												
10.	Discuss the architecture of Multi-layer perceptron with diagram.		2	2,4	2																																												
11.	Apply K Nearest Neighbor (K-NN) classifier to predict the diabetic patient with the given features BMI, age. Consider Test data : BMI=43.6, age=40, Sugar=? And assume k=3. <table><tr><th>Sl.NO</th><th>BMI</th><th>AGE</th><th>SUGAR</th></tr><tr><td>1</td><td>33.6</td><td>50</td><td>1</td></tr><tr><td>2</td><td>26.6</td><td>30</td><td>0</td></tr><tr><td>3</td><td>23.4</td><td>40</td><td>0</td></tr><tr><td>4</td><td>43.1</td><td>67</td><td>0</td></tr><tr><td>5</td><td>35.3</td><td>23</td><td>1</td></tr><tr><td>6</td><td>35.9</td><td>67</td><td>1</td></tr><tr><td>7</td><td>36.7</td><td>45</td><td>1</td></tr><tr><td>8</td><td>25.7</td><td>46</td><td>0</td></tr><tr><td>9</td><td>23.3</td><td>29</td><td>0</td></tr><tr><td>10</td><td>31</td><td>56</td><td>1</td></tr></table>	Sl.NO	BMI	AGE	SUGAR	1	33.6	50	1	2	26.6	30	0	3	23.4	40	0	4	43.1	67	0	5	35.3	23	1	6	35.9	67	1	7	36.7	45	1	8	25.7	46	0	9	23.3	29	0	10	31	56	1		2	2,4	3
Sl.NO	BMI	AGE	SUGAR																																														
1	33.6	50	1																																														
2	26.6	30	0																																														
3	23.4	40	0																																														
4	43.1	67	0																																														
5	35.3	23	1																																														
6	35.9	67	1																																														
7	36.7	45	1																																														
8	25.7	46	0																																														
9	23.3	29	0																																														
10	31	56	1																																														
12.	Explain the challenges of Unsupervised learning with examples.		3	1,4	2																																												
13.	Discuss the agglomerative and divisive approaches of clustering with examples.		3	1,4	2																																												
14.	Narrate your discussion on Credit Assignment Problem in Reinforcement Learning.		4	2,4	2																																												

15.	Explain how Deep Q-Learning is used to train an agent to play a game like Pacman.		4	2,4	2																																																																																										
16.	Write the steps of K - Means clustering algorithm with simple demonstration.		3	1,4	2																																																																																										
Section - C																																																																																															
Answer any FIVE of the following questions carrying 10 marks each		5x10=50																																																																																													
17.	Discuss the following data preprocessing tasks with examples: i) Data Cleaning ii) Data Integration iii) Data Transformation iv) Data Reduction		1	1,3	2																																																																																										
18.	Provide the architecture of Convolution Neural Network. Explain it with the components.		2	2,4	2																																																																																										
19.	Explain the ID3 algorithm. Given the dataset below, construct a decision tree using the ID3 algorithm. <table border="1"> <thead> <tr> <th>RID</th><th>age</th><th>income</th><th>student</th><th>credit_rating</th><th>Class: buys_computer</th></tr> </thead> <tbody> <tr><td>1</td><td>youth</td><td>high</td><td>no</td><td>fair</td><td>no</td></tr> <tr><td>2</td><td>youth</td><td>high</td><td>no</td><td>excellent</td><td>no</td></tr> <tr><td>3</td><td>middle_aged</td><td>high</td><td>no</td><td>fair</td><td>yes</td></tr> <tr><td>4</td><td>senior</td><td>medium</td><td>no</td><td>fair</td><td>yes</td></tr> <tr><td>5</td><td>senior</td><td>low</td><td>yes</td><td>fair</td><td>yes</td></tr> <tr><td>6</td><td>senior</td><td>low</td><td>yes</td><td>excellent</td><td>no</td></tr> <tr><td>7</td><td>middle_aged</td><td>low</td><td>yes</td><td>excellent</td><td>yes</td></tr> <tr><td>8</td><td>youth</td><td>medium</td><td>no</td><td>fair</td><td>no</td></tr> <tr><td>9</td><td>youth</td><td>low</td><td>yes</td><td>fair</td><td>yes</td></tr> <tr><td>10</td><td>senior</td><td>medium</td><td>yes</td><td>fair</td><td>yes</td></tr> <tr><td>11</td><td>youth</td><td>medium</td><td>yes</td><td>excellent</td><td>yes</td></tr> <tr><td>12</td><td>middle_aged</td><td>medium</td><td>no</td><td>excellent</td><td>yes</td></tr> <tr><td>13</td><td>middle_aged</td><td>high</td><td>yes</td><td>fair</td><td>yes</td></tr> <tr><td>14</td><td>senior</td><td>medium</td><td>no</td><td>excellent</td><td>no</td></tr> </tbody> </table>	RID	age	income	student	credit_rating	Class: buys_computer	1	youth	high	no	fair	no	2	youth	high	no	excellent	no	3	middle_aged	high	no	fair	yes	4	senior	medium	no	fair	yes	5	senior	low	yes	fair	yes	6	senior	low	yes	excellent	no	7	middle_aged	low	yes	excellent	yes	8	youth	medium	no	fair	no	9	youth	low	yes	fair	yes	10	senior	medium	yes	fair	yes	11	youth	medium	yes	excellent	yes	12	middle_aged	medium	no	excellent	yes	13	middle_aged	high	yes	fair	yes	14	senior	medium	no	excellent	no		2	2,4	3
RID	age	income	student	credit_rating	Class: buys_computer																																																																																										
1	youth	high	no	fair	no																																																																																										
2	youth	high	no	excellent	no																																																																																										
3	middle_aged	high	no	fair	yes																																																																																										
4	senior	medium	no	fair	yes																																																																																										
5	senior	low	yes	fair	yes																																																																																										
6	senior	low	yes	excellent	no																																																																																										
7	middle_aged	low	yes	excellent	yes																																																																																										
8	youth	medium	no	fair	no																																																																																										
9	youth	low	yes	fair	yes																																																																																										
10	senior	medium	yes	fair	yes																																																																																										
11	youth	medium	yes	excellent	yes																																																																																										
12	middle_aged	medium	no	excellent	yes																																																																																										
13	middle_aged	high	yes	fair	yes																																																																																										
14	senior	medium	no	excellent	no																																																																																										
20.	What is Principal Component Analysis (PCA)? Explain the steps involved in PCA with mathematical formulation.		3	1,4	2																																																																																										
21.	Describe Markov Decision Process and its role in reinforcement learning.		4	2,4	2																																																																																										
22.	Explain the DBSCAN algorithm in detail. Justify its need for the quality of clusters with example.		3	1,4	2																																																																																										

2nd Semester MCA Semester End Examination March 2026

Course Title: Object Oriented Programming using Java

Course Code: M23DE0202 - 48623202

Duration: 3 Hours

Max. Marks: 100

Note:

1. Answer the required number of questions from **Section A to C**.
2. Verify and ensure that question paper is completely printed before answering the question paper.
3. Any queries/discrepancies regarding the question paper, must be brought to the notice of the invigilator.
4. **Students must check the course title and course code before answering the question paper.**

Section - A		Marks	CO	PO	BL
Answer any FIVE of the following questions carrying 2 marks each		5x2=10			
1.	Mention any two features of Java that make it secure.		1	1	1
2.	Write any four important methods of ArrayList.		2	2	2
3.	What is the role of the Class Loader in Java Virtual Machine (JVM)?		1	3	2
4.	List the various types of inheritance supported by java.		3	3	1
5.	Write the importance of Java Database Connectivity (JDBC) in Java programming.		4	4	1
6.	List the features of Swing components.		4	2	2
Section - B					
Answer any EIGHT of the following questions carrying 5 marks each		8x5=40			
7.	Discuss the different types of tokens in Java with suitable examples.		1	4	2
8.	Explain super keyword in Java with examples for method overriding.		1	3	2
9.	Differentiate method overriding and method overloading in Java? Provide suitable examples to support your answer.		2	2	2
10.	Compare ArrayList and LinkedList in Java. Describe their use cases.		2	3	3
11.	How do lambda expressions improve the readability of code? Provide an example of using a lambda expression in a functional interface.		2	4	2
12.	Demonstrate how user-defined packages are created and used in Java.		3	4	2
13.	What are the features of interfaces in Java? Explain with an example how a class can implement multiple interfaces.		3	3	3
14.	Describe the use of JButton, JLabel, and JTextField in a Java Swing application.		4	2	2
15.	Differentiate Swing and AWT in Java? Explain with examples.		4	2	2
16.	Discuss the role of try, catch, and finally blocks in exception handling.		3	4	3
Section - C					
Answer any FIVE of the following questions carrying 10 marks each		5x10=50			
17.	Write a Java program that demonstrates implicit type conversion and explicit type casting, explaining the output.		1	3	3
18.	Elaborate the concept of autoboxing and unboxing with examples. How do wrapper classes simplify the conversion between primitive types and objects?		2	3	2
19.	Write a Java program to implement the following: Create a class Account with- - Attributes: accountNumber, balance. - Methods to:		1	4	3

	i. Deposit money: deposit(double amount). ii. Withdraw money: withdraw(double amount). iii. Display balance: displayBalance(). Demonstrate the functionality of the Account class by creating an instance and performing various operations.				
20.	Describe the process of multithreading in Java. What are the methods used to control thread execution?		3	2	2
21.	Discuss the various types of JDBC drivers and their use cases, including the advantages and disadvantages of each type.		4	2	3
22.	Write a program that demonstrates the usage of event listeners such as ActionListener for handling key press and button clicks.		4	3	3

2nd Semester MCA Semester End Examination March 2026

Course Title: Computer Networks

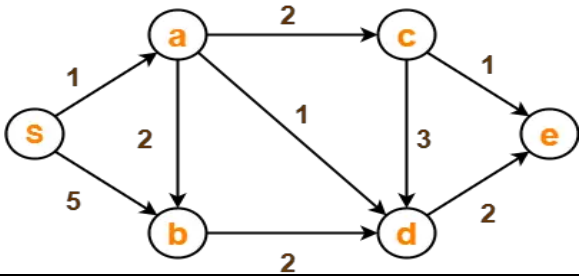
Course Code: M23DE0203 - 48623203

Duration: 3 Hours

Max. Marks: 100

Note:

1. Answer the required number of questions from **Section A to C**.
2. Verify and ensure that question paper is completely printed before answering the question paper.
3. Any queries/discrepancies regarding the question paper, must be brought to the notice of the invigilator.
4. **Students must check the course title and course code before answering the question paper.**

Section - A		Marks	CO	PO	BL
Answer any FIVE of the following questions carrying 2 marks each		5x2=10			
1.	What is the role of network software in data communication?		1	1	1
2.	Mention two features of sliding window protocols.		2	2	1
3.	Write any two differences between OSI and TCP/IP reference models.		1	2	2
4.	State the optimality principle in routing.		3	2	1
5.	Mention two uses of SMTP.		4	1	1
6.	Write the main difference between Distance Vector and Link State routing.		3	2	2
Section - B					
Answer any EIGHT of the following questions carrying 5 marks each		8x5=40			
7.	Give the layered architecture of the OSI model. Mention the function of each layer.		1	3	2
8.	Explain the causes of transmission impairment and methods to overcome them.		2	2	2
9.	Explain any five design issues of the Data Link Layer.		2	2	2
10.	A dataword 101100 is to be transmitted using CRC with the generator polynomial $G(x) = x^3 + x + 1$. Find the codeword to be transmitted. Show all steps of binary division.		2	3,4	3
11.	Describe the structure and working of the Public Switched Telephone Network (PSTN).		1	2,3	2
12.	Discuss Quality of Service (QoS) parameters in IP networks.		3	3,4	4
13.	Using Dijkstra's Algorithm, find the shortest distance from source vertex 'S' to remaining vertices in the following graph. 		3	3	3
14.	Explain the structure and function of the Domain Name System (DNS) in detail.		4	2,3	2,3
15.	Discuss the architectural overview of the World Wide Web and explain dynamic web document generation.		4	2,3	3
16.	Compare and contrast UDP and TCP .		4	3	4

Section - C				
Answer any FIVE of the following questions carrying 10 marks each		5x10=50		
17.	Present a summary of the various types of transmission media. Provide examples for each. Mention their advantages and disadvantages of each.		1	3
18.	Explain the steps involved in Hamming Code error detection and correction technique. Given a 4-bit data 1011, construct the 7-bit Hamming Code.		2	4,5
19.	Discuss the assumptions, working, and limitations of elementary data link protocols. Illustrate with examples.		3	2
20.	Explain the Distance Vector Routing algorithm in detail and discuss the count to infinity problem with a solution approach.		3	3
21.	Describe the functionalities of SMTP and HTTP. Analyze how each protocol operates and their roles in data communication over the internet.		4	2
22.	Explain the working of FTP in detail with control and data connection flow.		4	3

2nd Semester MCA Semester End Examination March 2026
Course Title: Web Application Development Framework
Course Code: M23DE0204 - 48623204

Duration: 3 Hours

Max. Marks: 100

Note:

1. Answer the required number of questions from **Section A to C**.
2. Verify and ensure that question paper is completely printed before answering the question paper.
3. Any queries/discrepancies regarding the question paper must be brought to the notice of the invigilator.
4. **Students must check the course title and course code before answering the question paper.**

Section - A		Marks	CO	PO	BL
Answer any FIVE of the following questions carrying 2 marks each		5x2=10			
1.	How do you create a hyperlink in HTML? Illustrate with an example.		1		2
2.	List any four features of JavaScript.		2		1
3.	Write the syntax to include CSS in a HTML document.		1		1
4.	Define XML namespace with an example.		3		1
5.	Define virtual DOM in React.		4		1
6.	Give an example of a JSON object with two key-value pairs.		3		2
Section - B					
Answer any EIGHT of the following questions carrying 5 marks each		8x5=40			
7.	Apply HTML form elements to create a book details form that collects information such as book title, ISBN number, author name, type of book, and price using at least four input fields.		1		3
8.	Discuss various CSS selectors with suitable examples.		1		2
9.	Apply different JavaScript array methods in a program to demonstrate their working with relevant examples.		2		3
10.	Develop a JavaScript program that uses a function to generate Fibonacci sequence up to 'n' terms. Show the output.		2		3
11.	Explain CSS box model and show its application with a suitable program.		1		2
12.	Discuss XML schema and its types with suitable examples.		3		2
13.	Compare XML and JSON. Construct a JSON object to represent employee details (EmployeeID, Name, Department, Salary).		3		3
14.	Explain the architecture and key features of AngularJS and ReactJS.		4		2
15.	Develop an AngularJS program to demonstrate two-way data binding with a text input field.		4		3
16.	How CSS can be implemented in XML? Illustrate with an example.		3		3
Section - C					
Answer FIVE of the following questions carrying 10 marks each		5x10=50			
17.	a) Design an HTML program to create the following table:		1		4

	<table><tr><th rowspan="2"></th><th colspan="2">Average</th><th rowspan="2">Red eyes</th></tr><tr><th>height</th><th>weight</th></tr><tr><td>Males</td><td>1.9</td><td>0.003</td><td>40%</td></tr><tr><td>Females</td><td>1.7</td><td>0.002</td><td>43%</td></tr></table>		Average		Red eyes	height	weight	Males	1.9	0.003	40%	Females	1.7	0.002	43%				2
	Average		Red eyes																
	height	weight																	
Males	1.9	0.003	40%																
Females	1.7	0.002	43%																
	b) Discuss different types of text styling supported in CSS.																		
18.	a) Illustrate date and time functions in JavaScript using suitable programming examples. b) Construct a JavaScript program to check if a given number is prime or not.		2		3 3														
19.	a) Write a CSS program to demonstrate how inline, internal, and external styles interact when applied to the same element. b) Illustrate various looping control structures supported in JavaScript using suitable programming examples.		1		3 3														
20.	Write a JSON array to store product details (product ID, name, price, availability) and construct a JavaScript program to display them in tabular format on a webpage.		3		4														
21.	a) Implement event handling in AngularJS by creating a program where a button click changes the background color of a webpage. b) Develop an AngularJS application that demonstrates two-way data binding using a text box and a paragraph.		4		4 4														
22.	a) Discuss the significance of XPath and XQuery. b) Write a XQuery using FLOWR expressions to retrieve all employees with a salary greater than 40,000 from an XML employee database and display their names and departments in sorted order.		3		3 3														

2nd Semester MCA Semester End Examination March 2026

Course Title: Artificial Intelligence

Course Code: M23DES221 - 48623210

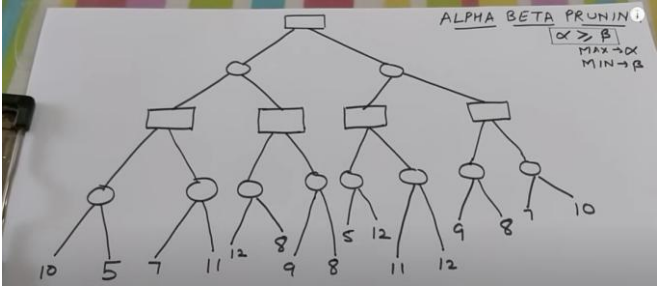
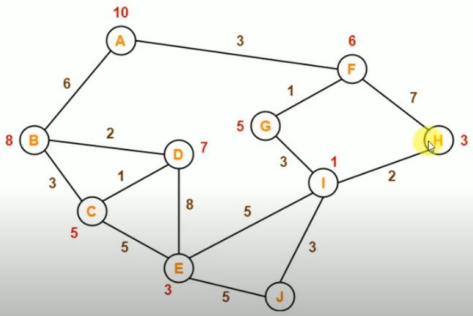
Duration: 3 Hours

Max. Marks: 100

Note:

1. Answer the required number of questions from **Section A to C**.
2. Verify and ensure that question paper is completely printed before answering the question paper.
3. Any queries/discrepancies regarding the question paper, must be brought to the notice of the invigilator.
4. **Students must check the course title and course code before answering the question paper.**

Section - A		Marks	CO	PO	BL
Answer any FIVE of the following questions carrying 2 marks each		5x2=10			
1.	What is Artificial Intelligence?		1	1	2
2.	Mention different types of Knowledge.		2	2	2
3.	List any two differences between Breadth First Search(BFS) and Depth First Search (DFS).		1	3	1
4.	What is planning?		3	1	2
5.	Define fuzzy logic.		4	4	2
6.	Mention any two real time Applications of Natural Language Processing (NLP).		4	1	1
Section - B					
Answer any EIGHT of the following questions carrying 5 marks each		8x5=40			
7.	Discuss the applications of AI		1	1	3
8.	Illustrate the steps to reach the goal from the start using Hill Climbing algorithm to the following blocks world. Start A D C B Goal D C B A Blocks World		1	3	4
9.	Converting sentence to first-order logic a) Every cat likes to sleep b) There is a dog that chases the mailman. c) No student failed the exam d) If it's sunny, then John goes for a walk e) Every parent loves their child.		2	2	4
10.	Discuss the logical connective and truth table in proposition logic.		2	3	3
11.	Explain default reasoning and minimalist reasoning.		2	4	3
12.	Describe different types of planning.		3	3	3
13.	Discuss various robotic software architecture.		2	5	3
14.	Explain the Fuzzy logic control with example.		4	2	3
15.	Differentiate the expert systems from traditional programming systems in terms of knowledge representation and processing.		4	1	4
16.	Elaborate the components of NLP.		4	2	4

Section - C				
Answer any FIVE of the following questions carrying 10 marks each		5x10=50		
17.	What is Agent? Explain different types of agent with example.		1	2
18.	Perform Alpha-Beta Pruning for the following tree: 		2	3
	Give the final values for Alpha and Beta.			4
19.	Discuss the A* algorithm. Apply A* algorithm to find the optimal path and cost from initial state 'A' to the goal state 'J' in the following graph. 		1	4
20.	What is a Neural Network? Briefly explain learning in Neural networks.		3	3
21.	Explain the phases of NLP.		4	3
22.	What is learning? Discuss the types of learning with example.		3	4
